Intersection of Lines and Planes

1. Given the plane $2x - y + 3z + 5 = 0$, the normal vector is _____.

2. If two planes are parallel, then the normals are _____.

3. If two planes are not parallel, then the solution to the system of equations is a _____.

4. If a line is not parallel to a plane, then the system of equations has _____ solution(s).

5. If a line is coincident with a plane, then the intersection of the line and plane is _____.

6. If the normals of three planes are scalar multiples and there is no solution to the corresponding system, then a geometric description is _____.

7. If the normals of three distinct planes are not scalar multiples of one another, and the number of solutions is 0 then a geometric description of this system is _____.

8. If a line and a plane do not intersect, then the line is _____ the plane.

9. If a line is perpendicular to a plane, then the line is _____ to the normal of the plane.

10. If $3x - y + 2z + 1 = 0$ is coincident with $6x - 2y + 4z + k = 0$, then $k =$ _____.

11. If a line is perpendicular to $x + y - 2z = 4$, then a direction vector of the line is _____.

12. If two lines in 3-space do not intersect, then the lines are _____ or _____.

ANSWERS

1. $(2, -1, 3)$
2. parallel (i.e. scalar multiples)
3. line of intersection
4. 1
5. the line itself (an infinite number of solutions)
6. two coincident planes and the third parallel but distinct, or three parallel and distinct planes
7. three non-parallel planes that form a triangular prism
8. parallel and distinct from
9. parallel
10. 2
11. $(1, 1, -2)$
12. parallel and distinct; skew